



JAIPUR NATIONAL UNIVERSITY

End Semester/Year Examination Dec -2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech(Computer Science Engineering / AIML/CY/Data Science)
Semester/Year:	V/III
Paper Code:	BTECSCO503T22
Title of Paper:	Data Base Management System
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
 2. Section-A consists of seven questions and
 3. Section-B consists of six questions.
- Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	Define a Database Management System (DBMS). Differentiate between a Database and a DBMS with suitable examples.	4	1	1
Q.2	Explain the three-schema architecture of a database with a neat diagram. Discuss its advantages.	4	2	2
Q.3	Draw an ER Diagram for a Library Management System, identifying entities, attributes, and relationships.	4	4	3
Q.4	Analyze the difference between Tuple Relational Calculus and Domain Relational Calculus. Mention their	4	3	4

5	Q.4	Write SQL queries for the following based on a Company database:	10	1	1
6		(a) Retrieve employee names and salaries whose department is 'HR'.			
4		(b) List all employees earning more than the average salary.			
		(c) Display the department name and average salary using GROUP BY and HAVING. Define the use of nested queries and aggregate functions.			
2	Q.5	Evaluate the ACID properties of transactions. Discuss how concurrency control and recovery mechanisms ensure these properties in multi-user environments. Give real-world examples where deadlocks or rollbacks occur.	10	5	5
3	Q.6	Propose a backup and recovery plan for a banking database. Include periodic backup strategies, logging techniques (undo, redo, undo-redo), and steps for crash recovery. Justify your plan in terms of reliability and cost-efficiency.	10	6	6

computational capabilities.

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|-----|--|---|---|---|
| Q.5 | Evaluate the role of normalization in database design. Which normal form do you consider most effective and why? | 4 | 4 | 5 |
| Q.6 | Design a simple SQL schema for a Student-Department database including primary and foreign keys. | 4 | 5 | 6 |
| Q.7 | Compare lock-based and timestamp-based concurrency control methods used in transaction management. | 4 | 4 | 4 |

SECTION - B

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|-----|--|----|---|---|
| Q.1 | Explain the three-schema architecture of DBMS in detail. Discuss how this architecture supports data abstraction and independence. Support your answer with a neat diagram. | 10 | 2 | 2 |
| Q.2 | Design a university database using the ER model that includes Students, Courses, Faculty, and Departments. Specify all relevant attributes, keys, and relationships, and convert the ER diagram into relational schemas. | 10 | 3 | 3 |
| Q.3 | Differentiate between Relational Algebra and Relational Calculus. Illustrate at least three relational algebra operations (JOIN, DIVISION, and PROJECTION) using suitable examples. | 10 | 3 | 4 |

- Q.4 Write
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(b)

(c)

- Q.5 Ev
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